

29/7

D. Thangam

192524399

Stack, Queue and Linked list Questions:

1. Stack and Queue:

Stack operations:

- * Push (54) $\rightarrow$ [54]
- * Push (52) $\rightarrow$ [54, 52]
- * Pop () $\rightarrow$ removes 52
- * push (55) $\rightarrow$ [54, 55]
- * Push (62) $\rightarrow$ [54, 55, 62]
- * $\delta = \text{POP}() \rightarrow \delta = 62$

Queue operation:

- * enqueue (21) $\rightarrow$ [21]
- * enqueue (24) $\rightarrow$ [21, 24]
- * dequeue () $\rightarrow$ removes 21
- * enqueue (28) $\rightarrow$ [24, 28]
- * enqueue (32) $\rightarrow$ [24, 28, 32]
- * $q = \text{dequeue}() \rightarrow AV = 24$

$$\delta + AV = 86$$

2. Positive Expression:

Expression: $105 + 606 / * 8 -$

$$* 105 + = 15$$

$$* 606 / = 10$$

$$* 15 \times 10 = 150$$

$$* 150 - 8 = 142$$



Answer: 142

3. Postfix Expression

Expression: $8\ 2\ 3^{\wedge} / 2\ 3^* + 5\ 1^* -$

$$* 2^3 = 8$$

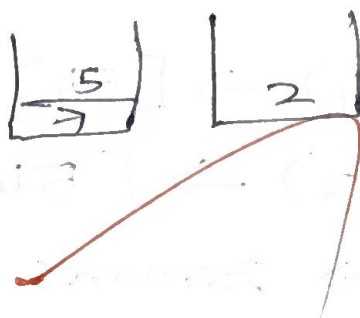
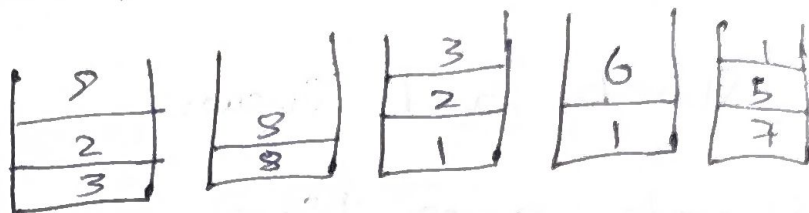
$$* 8 \div 8 = 1$$

$$* 2 \times 3 = 6$$

$$* 1 + 6 = 7$$

$$* 5 \times 1 = 5$$

$$* 7 - 5 = 2$$



Top two elements after the first $*$: 6, 1

Answer: 6 and 1

4. Prefix Infix to postfix:

Infix: $a + b \times c - d^{\wedge} e^{\wedge} f$

using precedence ($^{\wedge} > \times > +, -$)

and right associativity of $^{\wedge}$

Postfix: $abc + def^{\wedge\wedge} - *$

Answer: $abc + def^{\wedge\wedge} - *$

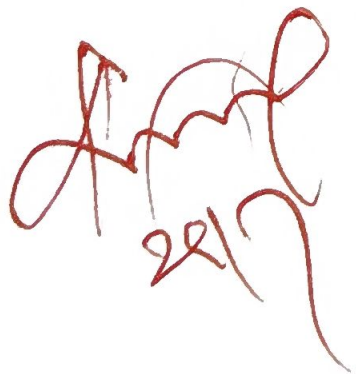
5. Stack pop sequence:

Operation:

- * Push (10)
- * Push (20)
- * Pop $\rightarrow$ 20
- * Push (10)
- * Push (20)
- * Pop $\rightarrow$ 20
- * Pop $\rightarrow$ 10
- * Push (20)
- * Pop $\rightarrow$ 20

Sequence : 20, 20, 10, 10, 20

Answer: 20, 20, 10, 10, 20


22/7